

MATHEW PIUS OLICKAL

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EDUCATION

San Jose State University MS Software Engineering Relevant Coursework: Enterprise Distributed Systems, Software Systems Engineering, Enterprise Software Platforms	San Jose, CA Aug 2026 - May 2028
Punjab Engineering College (PEC) BTech Computer Science and Engineering Relevant Coursework: Data Structures, Algorithms, Operating Systems, Computer Networks, DBMS, Machine Learning	Chandigarh, India 2018 - 2022

WORK EXPERIENCE

KPMG India – Digital Lighthouse <i>Software Engineer II (Client: Goldman Sachs)</i>	Bangalore, India March 2024 - July 2026
- Migrated reference-data sourcing for 5M+ daily regulatory transactions off legacy Sybase IQ, redesigning Apache Spark processing to support an authenticated upstream data platform. - Cut reference-data access latency by 95% (90 sec → 4 sec) by building reusable FINOS Legend (ALLOY) REST APIs with JWT-authenticated Snowflake and SingleStore sourcing. - Improved reference-data query response times by ~2x by developing optimized GraphQL queries and Java request/response models, reducing redundant retrieval through CTE-based execution. - Reduced production failure resolution time by ~30% by developing MCP servers that exposed internal Swagger APIs to an AI assistant, enabling data-lineage tracing to identify missing values and upstream refiner failures.	
<i>Software Engineer I (Client: Goldman Sachs)</i>	July 2022 - March 2024
- Reduced daily transaction fallout 90% (20,000 → 2,000) by redesigning Java validation and enrichment logic, contributing to ~\$2M in avoided compliance penalties. - Eliminated manual review of 100K+ records weekly by building a Java reconciliation application that automated SQL-based accuracy checks and scheduled execution through Autosys.	
BuildAR <i>Software Engineering Intern</i>	New Delhi, India Jan 2021 - May 2021
Reduced model-training data preparation time by 80% by building Python and FastAPI ingestion services for unstructured datasets and integrating backend APIs with JavaScript application components.	

PROJECTS

Vendor Independent Linux Server Health Monitor <i>Linux</i> github.com/mathewpius19/Vendor-Independent-Linux-Server-Health	<i>Python, Flask, Node.js, WebSockets, React, MongoDB, AWS EC2</i>
- Built a distributed Linux monitoring platform that deploys telemetry microservices over SSH and streams real-time system metrics through WebSockets to a centralized dashboard. - Developed a TensorFlow LSTM multivariate time-series model over live telemetry to forecast four server-health metrics and predict resource utilization trends and potential system failures.	
Reel-Finder github.com/mathewpius19/REEL-FINDER	<i>Python, Spring Boot, PostgreSQL, FastMCP, Ollama, FAISS, Sentence Transformers, Next.js</i>
- Designed a personalized recommendation pipeline using Sentence Transformers and FAISS vector search for semantic retrieval, re-ranking results using genre preferences, ratings, watch status, and interaction history. - Integrated an Ollama-powered AI agent with FastMCP tools to route natural-language queries to semantic search, invoking deterministic retrieval services only when movie recommendations are required.	

SOFTWARE PROFICIENCY

Programming Languages:	Java, Python, SQL, JavaScript, Bash, Object-Oriented Design
Backend & Distributed Systems:	Distributed Systems, Microservices, Spring Boot, Flask, FastAPI, Node.js, REST APIs, GraphQL, WebSockets
Databases & Data Engineering:	PostgreSQL, MongoDB, Snowflake, SingleStore, Sybase IQ, Apache Spark (PySpark)
Cloud, DevOps & Testing:	AWS (EC2), GitLab CI/CD, Docker, Git, Linux, Maven, Autosys, Integration Testing
AI & Developer Tools:	TensorFlow, LSTM, FAISS, Sentence Transformers, Vector Search, MCP, AI-Assisted Development

PROFESSIONAL CERTIFICATIONS

Databricks Certified Associate Developer for Apache Spark 3.0	Databricks	March 2023
Quantexa Professional Data Engineer Certification	Quantexa	Feb 2023